Dominik Szewczyk

Data Scientist · Machine Learning & AI

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PROFILE

Data Science and AI graduate (cum laude) who enjoys taking a project from start to finish. Across two internships and several client projects, I have handled the whole process myself, from data analysis and machine learning to building and delivering a working tool that meets the client's needs.

// 01 Experience

Data Scientist, Graduation Internship · KapiteinLabs

Dec 2025 - Jun 2026

- Built a predict-and-optimise machine learning system, delivered as a **no-code Streamlit dashboard** that engineers and clients use without writing code.
- Modelled a plasma reactor's ozone output from its physical settings (geometry, voltage, repetition rate, gas flow, pulse width) using **438 measurements** from TU Eindhoven.
- Benchmarked five models and trained a physics-informed neural network (PINN) as the best performer, reaching **R^2 = 0.988, MAE = 1.49 g/kWh, MAPE = 1.9%** on the test set.
- Built an inverse optimiser that recommends the settings needed to hit a target output, replacing slow one-off lab experiments with instant estimates, and documented it in a graduation thesis.

Data Scientist, Internship · De Rooy Transport & Logistics

Feb - Jun 2025

- Built a **multi-model pipeline** forecasting where, when, and how large unexpected pickups will be (over 82% of daily assignments), wired into the planners' PTV route software.
- Pickup occurrence: a Keras model over 5 km grid cells reaching **90.2% accuracy**, 88.9% precision, and 86.5% F1; location via random forest with an **18 m mean error** and 100% of predictions within 1 km; insertion time estimated with XGBoost.
- Engineered spatial, temporal, and holiday features for robustness at peak periods. Estimated impact of up to **€1,500, 1,500 km, and 945 kg CO_2 saved per day**. Featured in the LSB whitepaper.

Web Developer · Freelance

Feb 2026 - Present

- Built and deployed **10 live websites** for small-business clients, owning each project from design through to deployment and hosting.
- Worked directly with clients to scope requirements, iterate on design, and ship the final sites.

// 02 Education

Breda University of Applied Sciences · Data Science & Artificial Intelligence

2022 - 2026

Graduated cum laude. Coursework and projects across machine learning, computer vision, NLP, and MLOps, with two industry internships and a research-based graduation thesis.

II LO im. Andrzeja Frycza Modrzewskiego · Extended Mathematics

2019 - 2023

One of the top-ranked secondary schools in the region, with an extended mathematics profile.

// 03 Skills & Tools

MACHINE LEARNING & AI	Machine Learning, Deep Learning, Computer Vision, NLP & Transformers, Predictive Modeling, Feature Engineering, Reinforcement Learning
LANGUAGES & TOOLS	Python, scikit-learn, TensorFlow, Keras, XGBoost, FastAPI, Streamlit, Power BI, R, Git
CLOUD & MLOPS	Microsoft Azure, Azure ML, Docker, CI/CD pipelines, versioned assets
WAYS OF WORKING	Agile (Scrum), data visualisation, stakeholder communication, translating research into deployed tools
LANGUAGES	English (fluent) · Polish (native) · German (basic)
INTERESTS	Football, gym, chess (3rd category), half marathon (1:43)

// 04 Selected Projects

A selection of applied machine learning, computer vision, and analytics work across industry partners and university research. More detail and live demos at dominikszewczyk224180.github.io.

AI 3D Digital Twin & Object Removal

Antea Group & Kavel 10 · 2024 to 2025

Research on reconstructing urban environments from aerial and drone imagery for city planning. Benchmarked DeepFill v2, LaMa, and PowerPaint for generative object removal and U-Net, DeepLabV3, and FCN for semantic segmentation, then generated textured 3D meshes with Instant NeRF and NeRF Studio and built a VR walkthrough in Unreal Engine.

Computer Vision · NeRF · Semantic Segmentation · 3D Reconstruction

Plant Analysis with Computer Vision & Robotics

NPEC · 2023 to 2024

Segmented seeds, shoots, and roots from plant images with a U-Net and instance segmentation, localising root tips to within roughly 1 mm. Simulated a PID-controlled liquid-handling robot and applied reinforcement learning to dispense liquid precisely at the detected root tips.

Computer Vision · U-Net · Reinforcement Learning

ATHENA: MLOps for Plant Science

NPEC · 2024

Took research models to production for ATHENA, a Python package and cloud app for organ segmentation and landmark detection on high-resolution plant images. Built Azure data and training pipelines with versioned assets, containerised with Docker, and served the models via FastAPI. Work recognised publicly by NPEC.

Microsoft Azure · Docker · FastAPI · MLOps

Emotion Classification with NLP

Banijay Media · 2024

Detected emotions in video content by classifying text extracted from clip fragments. Compared Naive Bayes, logistic regression, HMM, CRF, RNN, and LSTM, then fine-tuned BERT to reach an F1 above 0.88 on the held-out test set.

NLP · Transformers (BERT) · Feature Engineering

Neighbourhood Safety & Crime Prediction

Municipality of Breda · 2023

A five-person Scrum project for the Municipality of Breda. Built a machine learning model that forecasts monthly crime rates per neighbourhood from historical data, paired with research into the drivers of neighbourhood safety, delivered through an interactive Streamlit dashboard for non-technical stakeholders.

Machine Learning · Forecasting · Streamlit

AI in Facility Management Studies

BUas · 2023

A mixed-method study into the impact of AI on facility management. Designed a Qualtrics survey, ran in-depth interviews, and analysed the data in R, then led the build of a Streamlit dashboard that was featured at the AI at BUas conference.

R · Mixed-method Research · Streamlit

Renewable Energy Across Europe (SDG 7)

SDG Hub, BUas · 2022

Analysed renewable energy consumption across Europe for Sustainable Development Goal 7, identifying countries that exceed the global average. Built an interactive Power BI dashboard and surfaced a strong upward trend with a correlation of 0.82 over time.

Power BI · EDA · Data Visualisation

Audience & Content Performance Analysis

Banijay Media · 2022 to 2023

Analysed content and ratings data to support media strategy. Cleaned and linked the data with a lookup table, then studied audience preferences, top-performing hosts, and the correlation between viewership and social media trends, giving the team a clearer picture of content performance.

Python · pandas · Time Series · Social Media Analytics